

Small-Angle Scattering by Fractal Systems

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Abstract

Fractal structures are characterized by self similarity within some spatial range. The mass distribution in a fractal object varies with a power D of the length R , smaller than the dimension d of the space. When the range of physical interest falls below 1000 Å, scattering techniques are the most appropriate way to study fractal structures and determine their fractal dimension D . Small-angle neutron scattering (SANS) is particularly useful when advantage can be taken of isotopic substitution. It is easy to show that the scattering law for a fractal object is given by $S(Q) \sim Q^{-D}$, where Q is the magnitude of the scattering vector. However, in practice some precautions must be taken because, near the limits of the fractal range, there are important deviations from this simple law. Some relations are derived which can be applied in relatively general situations, such as aggregation and gelation. The effects of polydispersity, important, in particular, in situations described by percolation models, are also shown.

Introduction

Since its introduction by Mandelbrot (1977), the concept of fractals has been used in many situations and systems, such as aggregates and gels. The fractal dimension very often summarizes and averages over very complex structural details. In physics this kind of description is useful if some physical property can be related to the fractal dimension. This is, for example, the case for aggregation, where different models, such as diffusion-limited aggregation (Witten & Sander, 1983), predict a well defined fractal dimension. However, the determination of the fractal dimension is not always easy, and some precautions must be taken when analysing experimental data.

Theory

For the purpose of this paper, a fractal object can be described by two properties. The first one is self similarity, which means that a detail of the fractal object is structurally identical to the whole or, in other

words, that the structure of the object is independent of the characteristic length scale of observation. It is clear that such a situation will break down at some scale in a real experiment. This means that, in the real world, an object can be considered as a fractal object and be described by a fractal dimension only within some spatial range, covering at least one decade. Near the borders of the fractal domain the ideal scattering law of the fractal object will no longer be obeyed and a specific treatment of the system cannot be avoided.

The second property which characterizes the fractal object is its fractal or Hausdorff dimension, D (Mandelbrot, 1977), defined as the exponent of the linear dimension R in the relation $M(R) \sim (R/r_0)^D$ where M represents the mass and r_0 is the gauge of measurement. The exponent D characterizes mass fractals. It is smaller than the dimension d of the space where the fractal object is embedded.

The concept of fractal dimension can easily be generalized to fractal surfaces. The fractal surface dimension D_s relates the measure of the surface and the linear length. Its limits are $d - 1$ and d .

In computer simulations, fractal objects are generated and their dimensions are determined from images in real space or their projections in a two-dimensional space, *e.g.* on photographs. If the simulation is performed in three dimensions, the fractal dimension is conserved in the projection if it is smaller than 2 (Weitz & Oliveria, 1984). Real experiments have been analysed as well in this way. The statistics of such a technique is in general poor. Conversely, scattering techniques, because of averaging in reciprocal space, are the appropriate way to determine a structure. This statement applies of course to fractal structures and some review papers have been devoted recently to the subject (Teixeira, 1986; Martin & Hurd, 1987).

Because the fractal structure never extends below the atomic separation, evaluations of fractal dimensions by scattering techniques concern small scattering vectors Q , say below 0.5 Å^{-1} . In a small- Q experiment the sample is analysed with a spatial resolution of the order of $2\pi/Q$. Typical values for Q in a small-angle neutron scattering (SANS) experiment extend from 10^{-3} to 1 Å^{-1} . Consequently, the fractal structure is easily observed by SANS if self similarity holds in some spatial range below 1000 Å.

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If the fractal behaviour extends over the entire experimental range the scattering law is particularly easy to derive. Actually, from the mass distribution $M(R) \sim R^D$, one obtains the density distribution $\rho \sim R^{D-d}$ (which goes to zero at infinite distances) and, by Fourier transform, the scattering law $S(Q) \sim Q^{-D}$. This gives the well known straight line with slope D in a logarithmic plot of $S(Q)$. Of course, this law applies only if one remains at values of Q large enough to be independent of the non-zero density of the sample at large scales. This law also breaks down for large values of Q when one analyses the sample over length scales small enough to be sensitive either to individual scatterers (the form factor) or to the surface (Porod region).

If the individual scatterers are relatively monodisperse, the scattered intensity $I(Q)$ can be decomposed as

$$I(Q) = \Phi P(Q) \bar{S}(Q). \quad (1)$$

In this expression, $\Phi = N/V_0$, where N is the number of scatterers and V_0 the volume of the sample. Φ is a number density expressed in cm^{-3} .

$P(Q)$ is a function of the form factor $F(Q)$ which will be defined below. It depends on the spatial distribution of the scattering lengths of the atoms constituting the sample. A precise determination of $P(Q)$ is not always possible, but simple expressions can be derived for some simple geometrical shapes. In principle, one must start from the coherent scattering length b_i of each individual atom i and evaluate, from their instantaneous positions $\mathbf{r}_i$, the density

$$\rho(\mathbf{r}) = \sum_i b_i \delta(\mathbf{r} - \mathbf{r}_i). \quad (2)$$

[Because the distribution $\delta(\mathbf{r} - \mathbf{r}_i)$ has the inverse dimensions of a volume, $\rho(\mathbf{r})$ is expressed in cm^{-2} .]

For practical purposes, in SANS the spatial resolution is low enough to allow some spatial averaging. In this context it is useful to introduce the notion of contrast in the following way. $\rho(\mathbf{r})$ is averaged over distances small compared with $1/Q$. The medium where the fractal object is immersed (often called the solvent) is in general well represented by a single number ρ_0 because its structure cannot be observed with the resolution of the SANS experiment. Very often the particle itself is supposed to have a uniform density. The contrast is given by $\rho(\mathbf{r}) - \rho_0$. One of the important advantages of SANS is that the contrast can very often be modified by isotopic substitution which has only minor effects on the structure. This is currently used with hydrogenated samples, owing to the large differences in the scattering lengths of hydrogen and deuterium.

The form factor $F(Q)$ is the Fourier transform of the contrast evaluated over the volume of the particle and $P(Q)$ is the square of the modulus of $F(Q)$ averaged

over the ensemble of particles:

$$P(Q) = \langle |F(Q)|^2 \rangle = \left\langle \left| \int_{\text{volume of the particle}} [\rho(\mathbf{r}) - \rho_0] \exp(i\mathbf{Q} \cdot \mathbf{r}) d\mathbf{r} \right|^2 \right\rangle \quad (3)$$

[$P(Q)$ is expressed in cm^2].

For some simple shapes $P(Q)$ is easily determined (Guinier & Fournet, 1955). For example, for uniform spheres of density ρ , volume V and radius r_0 ,

$$P(Q) = V^2(\rho - \rho_0)^2 \times [3 \sin(Qr_0) - Qr_0 \cos(Qr_0)/(Qr_0)^3]^2. \quad (4)$$

At large values of Q the scattered intensity becomes independent of the shape of the particles and is dominated by surface scattering. This is particularly true if a small polydispersity is present, as is always the case in real systems. In this domain, $P(Q)$ is given by the Porod law

$$P(Q) = 2\pi(\rho - \rho_0)^2 S/Q^4 \quad (5)$$

where S is the total surface. In this derivation, the density profile is supposed to vary sharply over distances smaller than $1/Q$. If the density profile varies less sharply, one obtains smoother Q dependences (Auvray, 1986).

The Porod law can be generalized (Bale & Schmidt, 1984) to fractal surfaces of dimension D_s . Writing

$$S = (r/r_0)^{D_s} \quad (6)$$

for the measurement of the fractal surface with dimension D_s , one obtains

$$P(Q) = \pi(\rho - \rho_0)^2 S \Gamma(5 - D_s) \times \sin[\pi(D_s - 1)/2] Q^{-(6 - D_s)}, \quad (7)$$

which gives (5) when $D_s = 2$.

Because $6 - D_s$ falls between 3 and 4, the Q dependence of the scattering law of a fractal surface is not easily distinguishable from that of a smooth interface.

$\bar{S}(Q)$ is an effective structure factor (Chen & Bendouch, 1985), given by

$$S(Q) = 1 + \frac{|\langle F(Q) \rangle|^2}{\langle |F(Q)|^2 \rangle} [S(Q) - 1]. \quad (8)$$

For a centrosymmetric particle $\bar{S}(Q) = S(Q)$. $S(Q)$ describes the spatial distribution of the individual scatterers. It is the Fourier transform of the pair correlation function, $g(r)$. [$g(r)$ is dimensionless.] $\Phi g(r)$ represents the probability of finding a particle at a distance r from a particle situated at the origin.

Explicitly, $S(Q)$ can be written as [see, for example, Squires (1978)]

$$S(Q) = 1 + \Phi \int [g(r) - 1] \exp(i\mathbf{Q} \cdot \mathbf{r}) d\mathbf{r}. \quad (9)$$

For isotropic systems

$$S(Q) = 1 + 4\pi\Phi \int_0^\infty [g(r) - 1]r^2 \sin(Qr)/Qr \, dr. \quad (10)$$

If the individual scatterers are sufficiently far apart (dilute system), $S(Q)$ reduces to unity and the total scattered intensity depends only on the form factor. Whenever possible, dilute systems are used to determine $P(Q)$.

Above some concentration, the individual scattering objects can no longer be seen as isolated. There will be at least a steric effect. In situations where there are, moreover, interparticle interactions, such as in systems containing charged particles, structural effects can appear even at extremely low concentrations, avoiding the direct determination of $P(Q)$. Methods have been derived to evaluate the structure factor in such situations (Guinier & Fournet, 1955).

The structure factor $S(Q)$ very often has a liquid-like shape with one or more oscillations, and goes necessarily to unity at large Q . It is unfortunately frequent practice, in the difficult interpretation of $S(Q)$, that a physical significance is attributed to the position of the first structural peak. Such a procedure is incorrect, because the structure factor is the Fourier transform of a pair correlation function. The first structural peak cannot in general be associated with some enlarged Bragg peak, as in a disordered solid material. At most, its position will give the order of magnitude of some characteristic distance in the sample. Because the measured Q range in a typical SANS experiment does not allow the evaluation of the Fourier integral, the determination of $g(r)$ must start from a plausible model able to reconstitute the measured scattering function.

For a fractal object, the structure factor $S(Q)$ can be derived in the following way. From the meaning of $g(r)$, the total number of particles within a sphere of radius r centred in a central particle is given (for $d = 3$) by

$$N(r) = \Phi \int_0^r g(r) 4\pi r^2 \, dr \quad (11)$$

or

$$dN(r) = \Phi g(r) 4\pi r^2 \, dr. \quad (12)$$

On the other hand, a fractal object is characterized by a spatial distribution of the individual scatterers given by

$$N(r) = (r/r_0)^D, \quad (13)$$

where r_0 is the gauge of the measurement, which has the magnitude of the characteristic dimension of each individual scatterer.

Differentiation of (13) and identification with (12) gives

$$\Phi g(r) = (D/4\pi r_0^D) r^{D-3}. \quad (14)$$

Because D is smaller than 3, $g(r)$ goes to zero at large r , as we have seen above. This is clearly unphysical. At some large scale, the sample will show a macroscopic density. A good knowledge of the sample allows in general a reasonable assumption for the large-scale behaviour of $g(r)$. We will adopt here a procedure first introduced by Sinha, Freltoft & Kjems (1984), which proved to be very useful, even if not absolutely general. It involves an expression of the form $\exp(-r/\xi)$, where ξ is a cut-off distance, to describe the behaviour of the pair correlation function at large distances. To derive the analytical form of $S(Q)$ within this assumption, one can use the general theory of liquids, where the uniform density is subtracted to avoid a divergence in the evaluation of $S(Q)$ (Chen & Teixeira, 1986; Teixeira, 1986). We then write

$$\Phi[g(r) - 1] = (D/4\pi r_0^D) r^{D-3} \exp(-r/\xi). \quad (15)$$

The introduction of the cut-off distance ξ is done in analogy with the Ornstein–Zernike treatment of critical phenomena (Stanley, 1971), which is represented by (15) with $D = 2$.

The meaning of ξ is only qualitative and has to be made precise in any particular situation. Generally speaking, it represents the characteristic distance above which the mass distribution in the sample is no longer described by the fractal law. In practice, it can represent the size of an aggregate or a correlation length in a disordered material.

Fourier transformation of (15) gives, in accordance with (10),

$$\begin{aligned} S(Q) &= 1 + \frac{D}{r_0^D} \int_0^\infty r^{D-1} \exp(-r/\xi) \frac{\sin(Qr)}{Qr} \, dr \\ &= 1 + \frac{1}{(Qr_0)^D} \frac{\Gamma(D-1)}{[1 + 1/(Q^2 \xi^2)]^{(D-1)/2}} \\ &\quad \times \sin[(D-1) \tan^{-1}(Q\xi)] \end{aligned} \quad (16)$$

where $\Gamma(x)$ is the gamma function of argument x .

This expression reduces of course to $S(Q) \sim Q^{-D}$ when $\xi^{-1} < Q < r_0^{-1}$. Expressions (1), (3) to (7) and (16) have been used with success to fit experimental data in many different situations. As an example, we give in Fig. 1 the scattered intensity from different samples of silica aerogel and their fit by the preceding equations (Courtens & Vacher, 1987).

The unity term in $S(Q)$ becomes dominant at large Q . Then, the scattered intensity $I(Q)$ is dominated by the form factor $P(Q)$ [see (1)]. By contrast, at small values of Q the intensity $I(Q)$ is dominated by $S(Q)$ and for $Q \ll \xi^{-1}$ has the generalized Guinier-type behaviour

$$\begin{aligned} \lim_{Q \rightarrow 0} S(Q) &= \Gamma(D+1)(\xi/r_0)^D \\ &\quad \times \{1 - [D(D+1)/6]Q^2 \xi^2\}. \end{aligned} \quad (17)$$

A comparison of this expression with Guinier's law shows that ξ is related to a generalized radius of gyration R_g by

$$R_g^2 = D(D+1)\xi^2/2.$$

In a general way, $S(Q)$ can be considered as the form factor of an ensemble of fractal objects distributed at a larger scale in the space.

Polydispersity

A particular situation is that of gelation processes described by percolation models (Stauffer, 1976; de Gennes, 1979), where a large polydispersity is present. During the gelation process of branched polymers, there is a very large distribution $n(s)$ of polymers of different sizes s . This distribution is characterized (Fig. 2) by two numbers: the average size, N_w , and the maximum cut-off size, N_z .

Within the percolation theory, N_w and N_z , near the percolation threshold, are associated with the two universal exponents, γ and σ .

$$N_w \sim \varepsilon^{-\gamma} \quad (18)$$

$$N_z \sim \varepsilon^{-1/\sigma} \quad (19)$$

where ε measures the distance to percolation.

The distribution $n(s)$ has been derived by Stauffer (1985) and depends on the exponent τ :

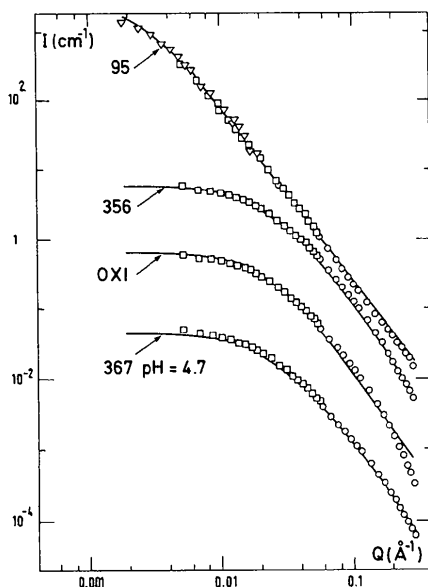


Fig. 1. Scattered intensity from several samples of silica aerogels. Numbers labelling the curves give the densities in g cm^{-3} and OXI means an oxidized sample with density equal to 356 g cm^{-3} . The sample labelled pH = 4.7 was prepared with acid catalysis. All the others are neutrally reacted samples. Solid lines represent fits with equations (1), (5) and (16). Each intensity is divided by ten compared with the previous one, starting from the top. (From Courtens & Vacher, 1987.)

$$n(s) = s^{-\tau} f_1(\varepsilon s^\sigma) = s^{-\tau} f_1(s/N_z), \quad (20)$$

where $f_1(x)$ is an exponentially decaying function which accounts for the cut off of the distribution at large s .

Near the percolation threshold, each polymer has a fractal structure with dimension D . The radius of gyration of each polymer can be associated with the cut-off parameter ξ . Consequently, one can write

$$s = \xi^D \quad (21)$$

where ξ is now the radius of gyration of the polymer of size s .

The pair correlation function of a single polymer containing s monomers (the individual scatterers) has the form of (15), where D is the fractal dimension of the polymer. In a more general way, one writes, instead of (15),

$$\begin{aligned} \Phi[g(r) - 1] &= (D/4\pi r_0^D) r^{D-3} f_2(-r/\xi) \\ &= (D/4\pi r_0^D) r^{D-3} f_2(-r/s^{1/D}), \end{aligned} \quad (22)$$

where we use (21). $f_2(-r/\xi)$ represents a decreasing function of r .

The scattering function of such a polymer is obtained, following the same steps as for the derivation of (16), as

$$\begin{aligned} S(s, Q) &= 1 + \frac{D}{r_0^D} \int_0^\infty r^{D-3} f_2(-r/s^{1/D}) \frac{\sin(Qr)}{Qr} dr \\ &= 1 + \frac{D}{(r_0 Q)^D} f_3(Q\xi) \\ &= 1 + s f_4(Q^D s) \end{aligned} \quad (23)$$

where $f_3(x)$ and $f_4(x)$ can be determined if $f_2(-r/\xi)$ is known.

The total scattering function at infinite dilution is now obtained (Martin & Ackerson, 1985; Bouchaud, Delsanti, Adam, Daoud & Durand, 1986) by summing all the contributions $S(s, Q)$ with their relative

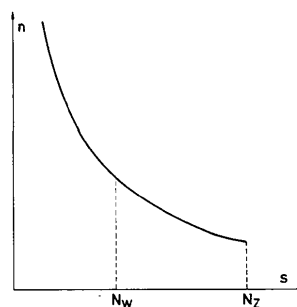


Fig. 2. Schematic representation of the number $n(s)$ of polymers with size s , near the percolation threshold. N_w and N_z are the average and the maximum size, respectively. This distribution is described by equation (20).

weights, $n(s)$, given by (20):

$$\begin{aligned} S(Q) - 1 &= \int ds sn(s) [S(s, Q) - 1] \\ &= \int ds s^2 s^{-\tau} f_1(s/N_z) f_4(Q^D s) \\ &= Q^{-(3-\tau)D} f_5(QR_z) \end{aligned} \quad (24)$$

where $R_z \sim N_z^{1/D}$. This result applies only for $\tau > 2$.

For large values of Q ($Q \gg R_z^{-1}$), $f_5(QR_z) \sim 1$. Comparing (16) and (24), one notices that, in the fractal region, the scattered intensity is now represented in a logarithmic plot by a straight line with a slope $(3-\tau)D$, instead of D . The theory of percolation predicts $D=2$ and $\tau=2.2$. In a recent experiment Bouchaud *et al.* (1986) have measured these exponents using fractionated samples containing only the polymers of large size. In the polydisperse sample, the dependence on Q of the scattered intensity gives $(3-\tau)D=1.59$ and, in the fractionated sample, D is evaluated directly giving $D=2$ (Fig. 3), in excellent agreement with the theory.

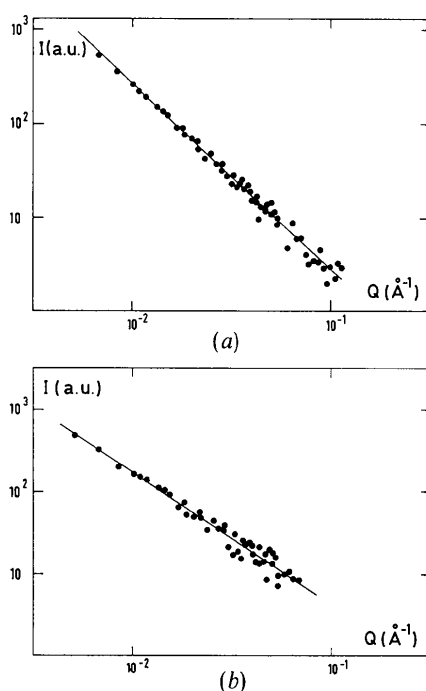


Fig. 3. (a) Scattered intensity by a fractionated sample containing only polymers with a large size. The slope gives $D=1.98$. (b) Scattered intensity by the dilute polydisperse sample. The slope gives $D(3-\tau)=1.59$. (From Bouchaud *et al.* 1986.)

These results show that fractal concepts are useful in interpreting complex situations in disordered materials and are the most convenient method for studying physical phenomena such as aggregation and gelation. Scattering techniques and SANS, in particular, appear to be the most appropriate technique for the determination of the fractal dimension. We have summarized some simple and relatively general expressions which can be derived to describe the intensity scattered by fractal objects.

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